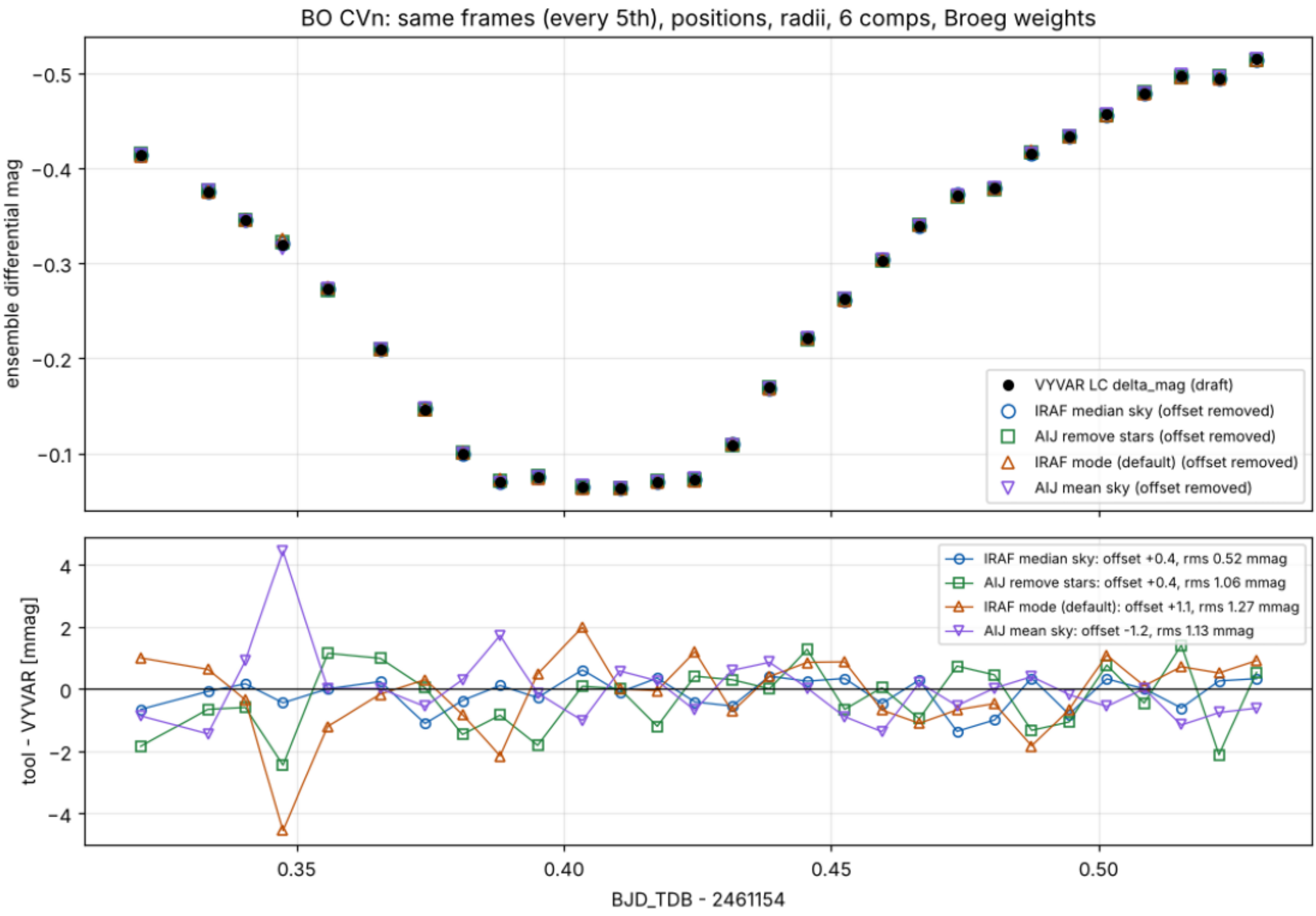
VYVAR cross-validation 2026-10-12: IRAF apphot / AstroImageJ vs VYVAR (differential LC, mmag)

Same pixels (aligned frames, every 5th), MASTERSTAR positions, VYVAR radii per frame (r = f* x FWHM_frame, annulus 2.7-5.2 FWHM),
the final comps and Broeg weights of each LC. 'VYVAR repro' = independent photutils recomputation vs the draft's mag_inst.

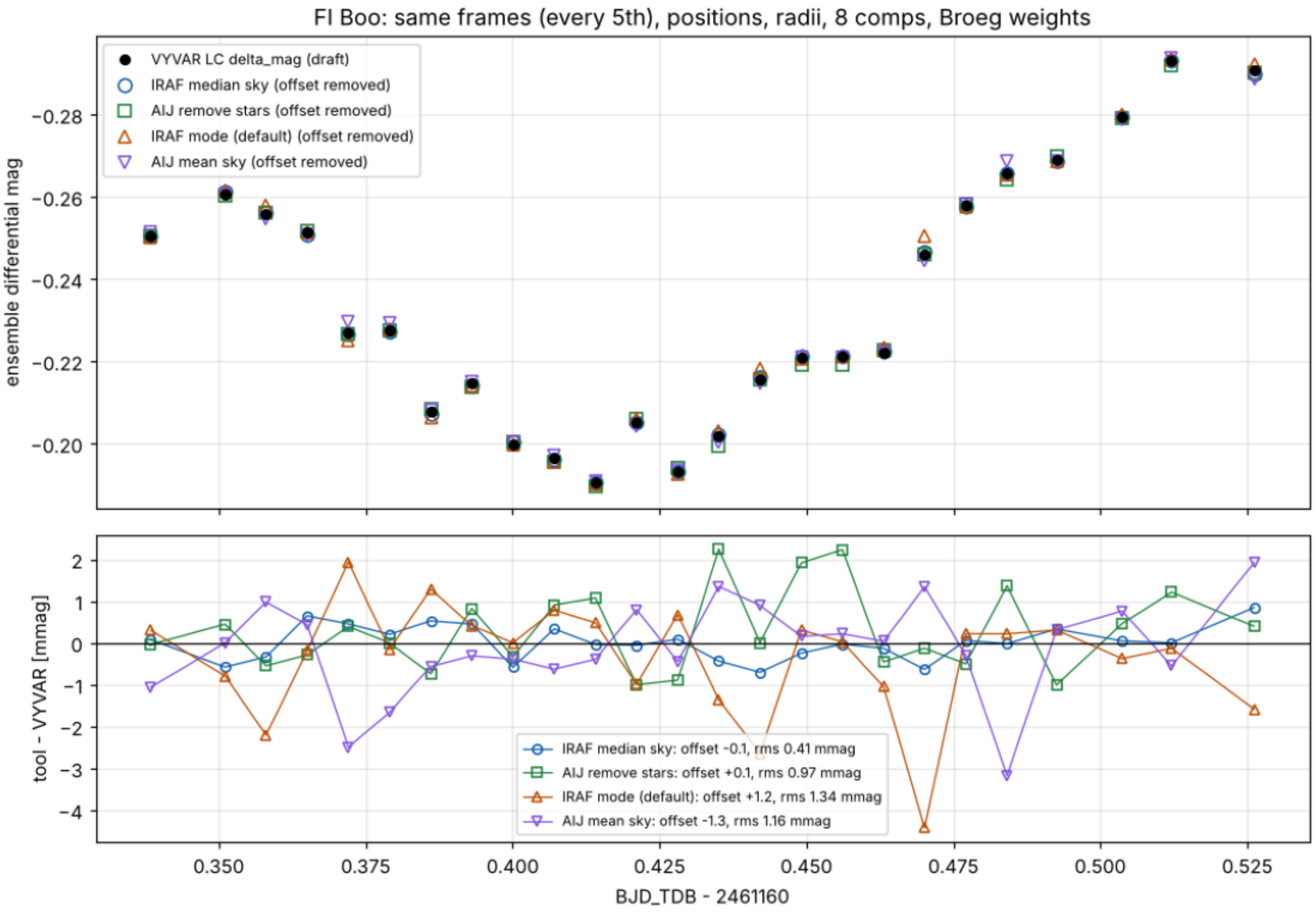
draft / setup	target	frames	comps	VYVAR repro max mmag	IRAF median offset	IRAF median rms	IRAF mode rms	AIJ rm-stars rms	AIJ mean sky offset	AIJ mean sky rms
525_BO_CVn	BO CVn	29	6	0.055	0.36	0.52	1.27	1.06	-1.21	1.13
526_FI_Boo	FI Boo	25	8	0.165	-0.15	0.41	1.34	0.97	-1.27	1.16
526_MN_Boo	MN Boo	27	8	0.181	-2.24	1.04	2.31	1.05	-7.88	1.5
526_g_120_4	S4 Cyg	18	8	0.202	-0.37	0.82	3.87	2.53	153.84	41.05
526_r_100_4	S4 Cyg	18	8	0.081	-0.29	0.92	2.49	2.06	41.92	19.79
527_V0842_Her	V0842 Her	27	8	0.169	-0.26	0.77	1.34	1.2	-21.75	6.81
528_CX_CVn	CX CVn	19	8	0.099	-0.17	1.06	1.54	0.5	-1.05	0.61

Reading: IRAF with the same sky estimator (median) reproduces VYVAR to 0.4-1.1 mmag rms (point errors 5-20 mmag).
Constant offsets (MN Boo -2.2 mmag) = IRAF partial-pixel area approximation at r = 2.7 px (area +0.4..1.6 % star by star).
IRAF mode / AIJ remove-stars differ only by the sky estimator (1-4 mmag rms). AIJ plain mean sky is biased by stars in the annulus
(S4 Cyg +154 mmag, V0842 Her -22 mmag) - VYVAR's median is robust to that.
S4 Cyg g/r: one comp lies 29 px from the chip edge; in frames with FWHM > 5.6 px its sky annulus leaves the chip
(VYVAR uses the finite part of the annulus; the stamps for the other tools were filled → repro 0.8 mmag).

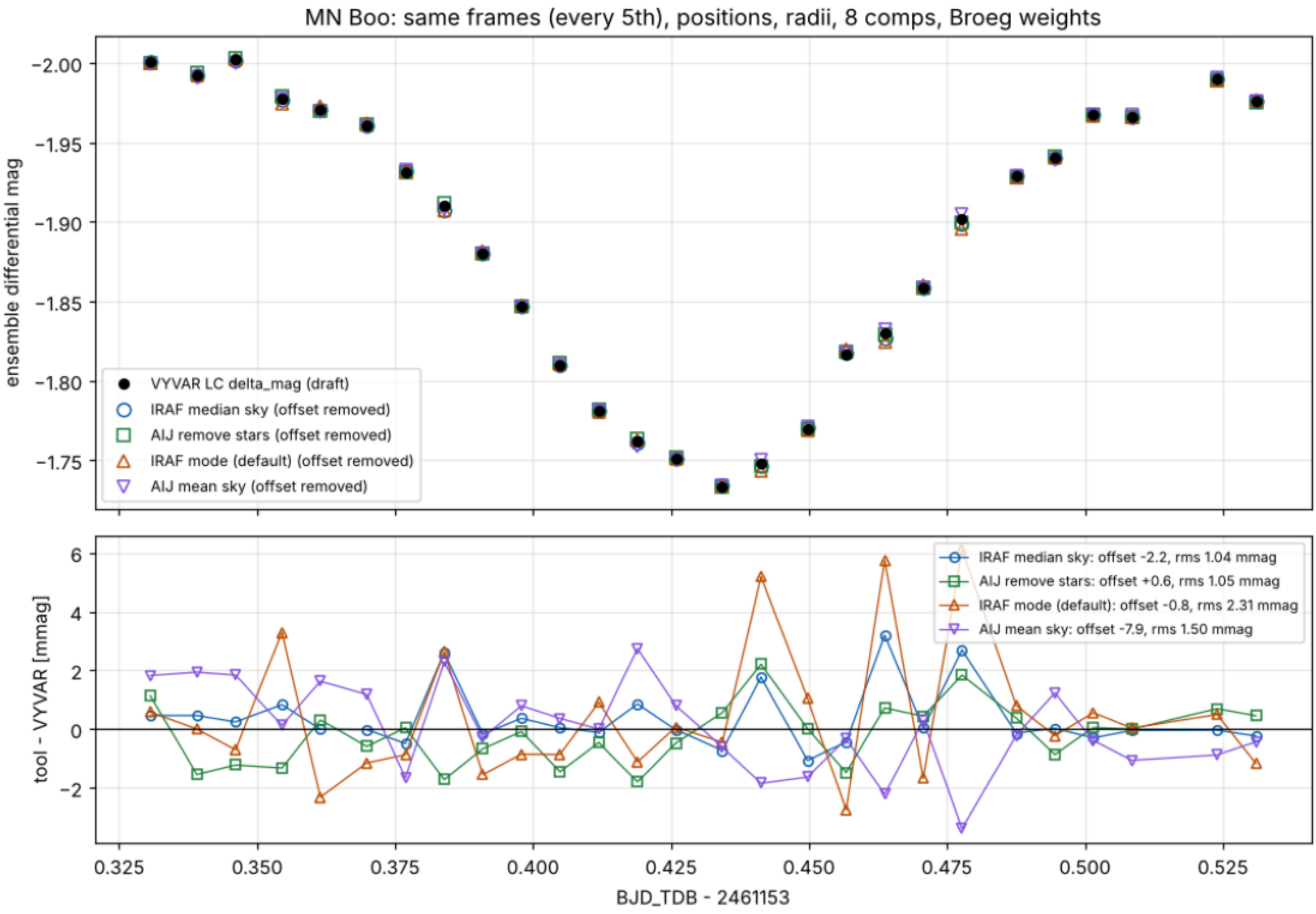
draft_000525_BO_CVn_NoFilter_60_2

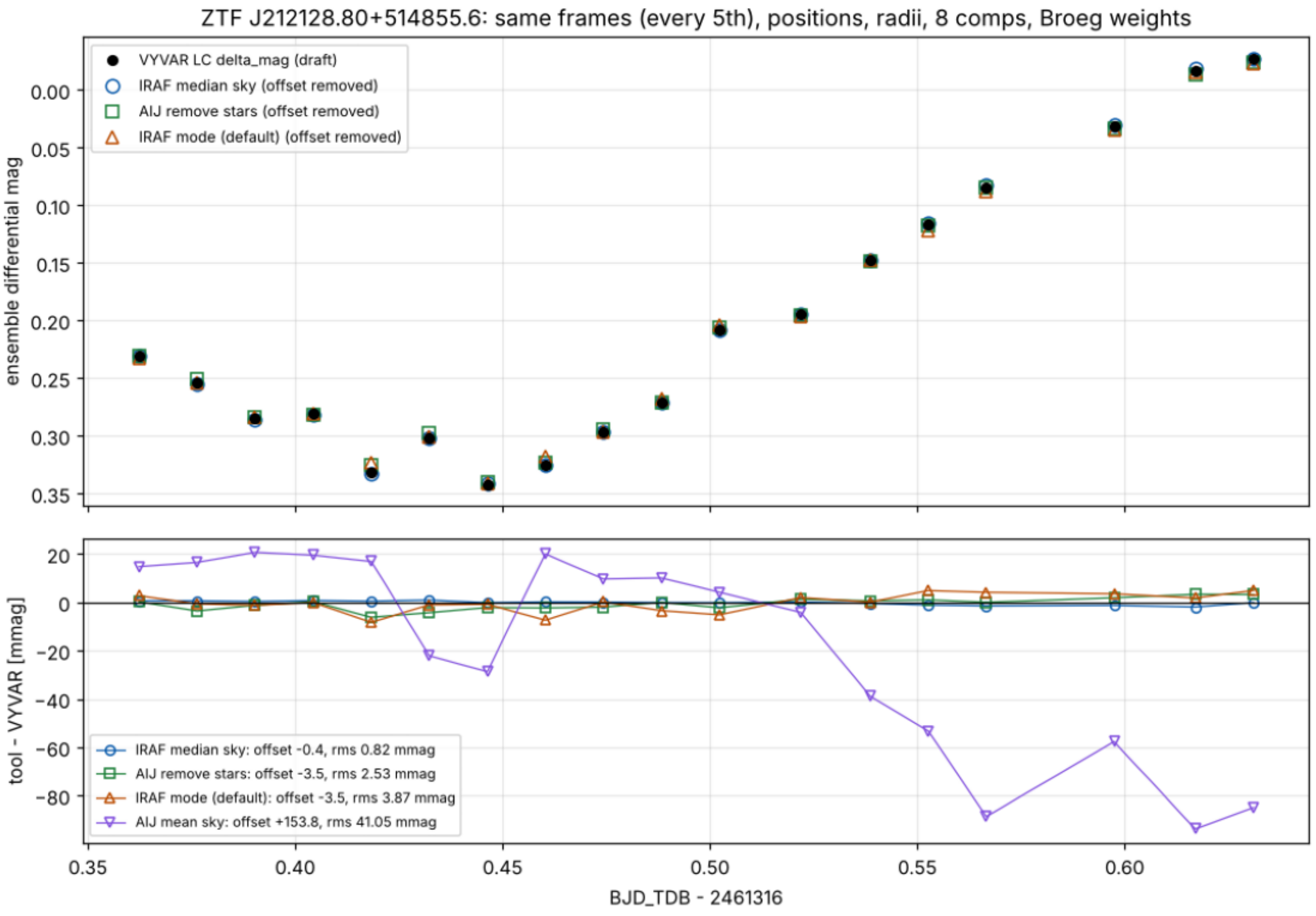


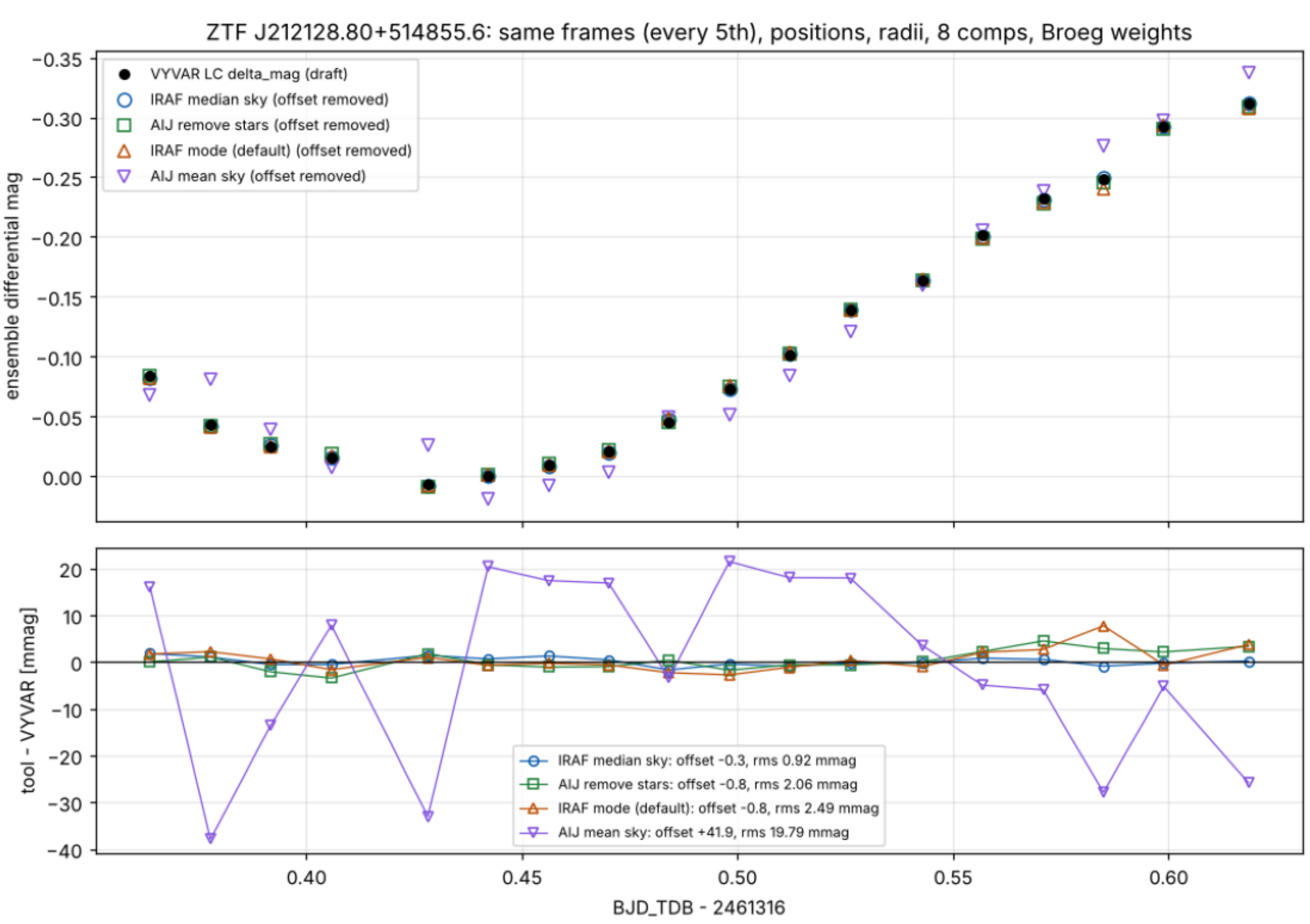
draft_000526_FI_Boo_NoFilter_60_2



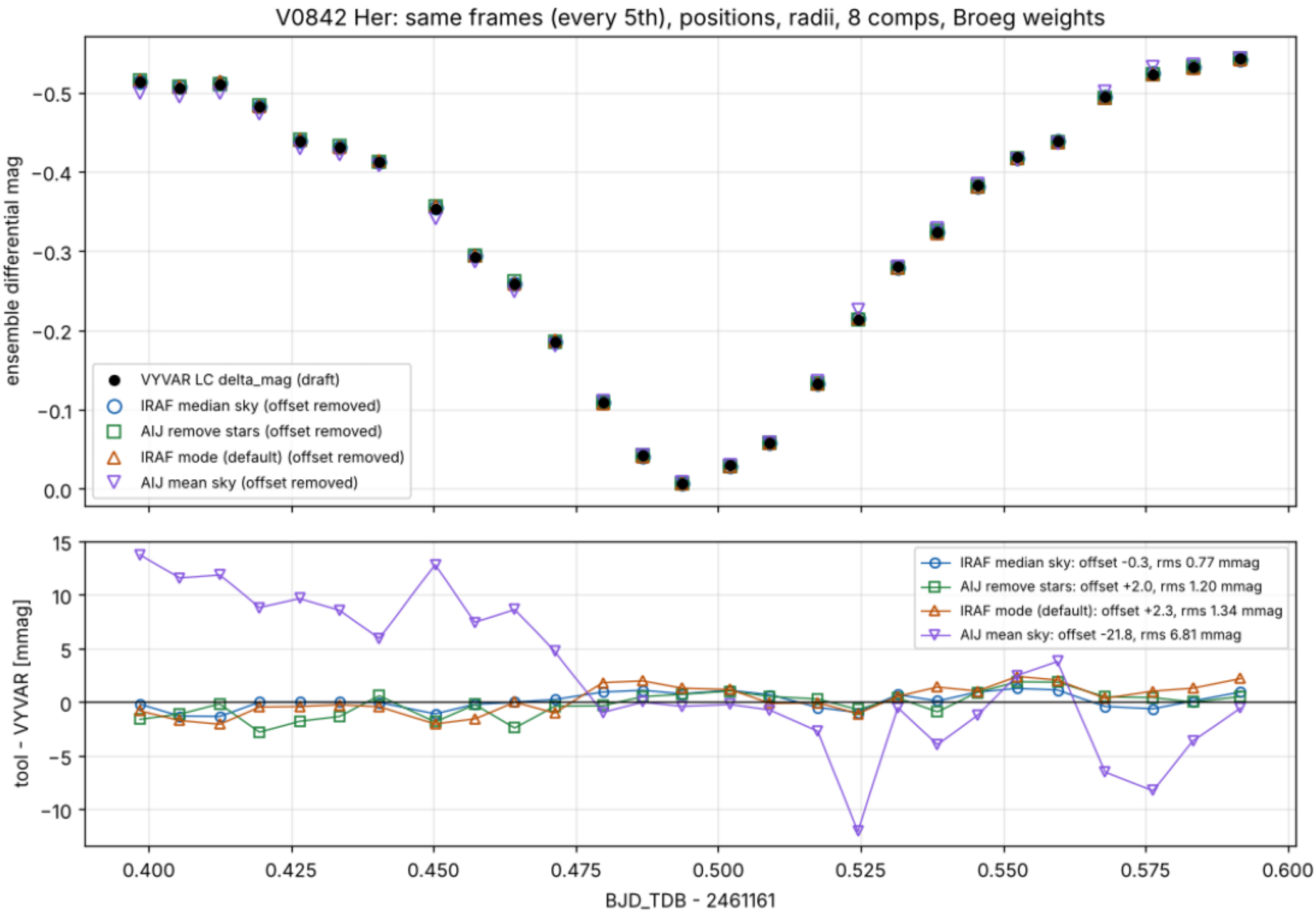
draft_000526_MN_Boo_NoFilter_60_2



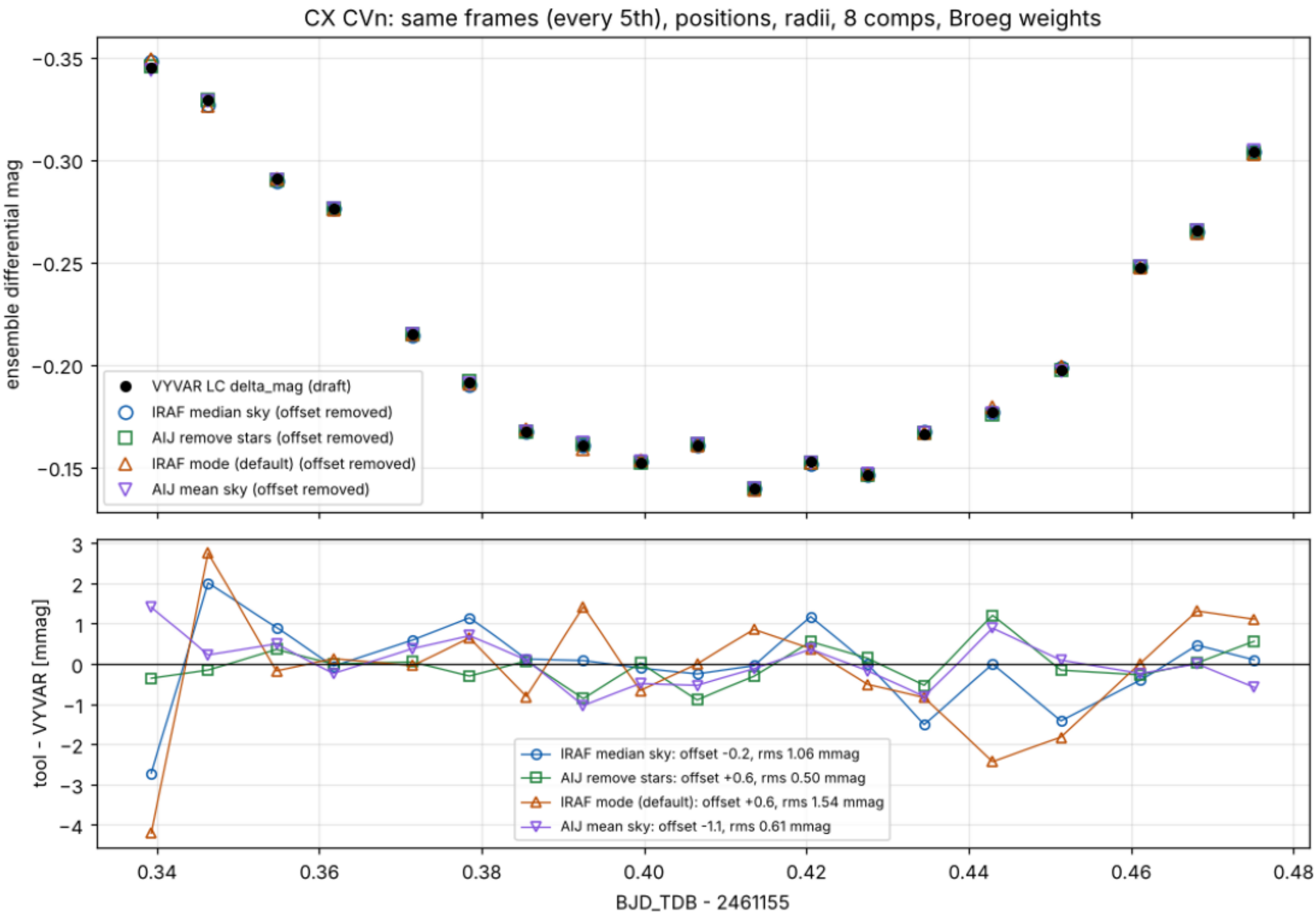

ZTF J212128.80+514855.6: same frames (every 5th), positions, radii, 8 comps, Broeg weights
VYVAR LC delta_mag (draft)
IRAF median sky (offset removed)
AIJ remove stars (offset removed)
IRAF mode (default) (offset removed)
ensemble differential mag
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
tool - VYVAR [mmag]
20
0
−20
−40
−60
−80
IRAF median sky: offset -0.4, rms 0.82 mmag
AIJ remove stars: offset -3.5, rms 2.53 mmag
IRAF mode (default): offset -3.5, rms 3.87 mmag
AIJ mean sky: offset +153.8, rms 41.05 mmag
0.35
0.40
0.45
0.50
0.55
0.60
BJD_TDB - 2461316


ZTF J212128.80+514855.6: same frames (every 5th), positions, radii, 8 comps, Broeg weights
VYVAR LC delta_mag (draft)
IRAF median sky (offset removed)
AIJ remove stars (offset removed)
IRAF mode (default) (offset removed)
AIJ mean sky (offset removed)
ensemble differential mag
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
tool - VYVAR [mmag]
20
10
0
−10
−20
−30
−40
IRAF median sky: offset -0.3, rms 0.92 mmag
AIJ remove stars: offset -0.8, rms 2.06 mmag
IRAF mode (default): offset -0.8, rms 2.49 mmag
AIJ mean sky: offset +41.9, rms 19.79 mmag
0.40
0.45
0.50
0.55
0.60
BJD_TDB - 2461316

draft_000527_V0842_Her_NoFilter_60_2



draft_000528_CX_CVn_NoFilter_60_2



Muniwin (colleague) vs VYVAR, S4 Cyg = ZTF J212128.80+514855.6

